



Format of a research paper

BY

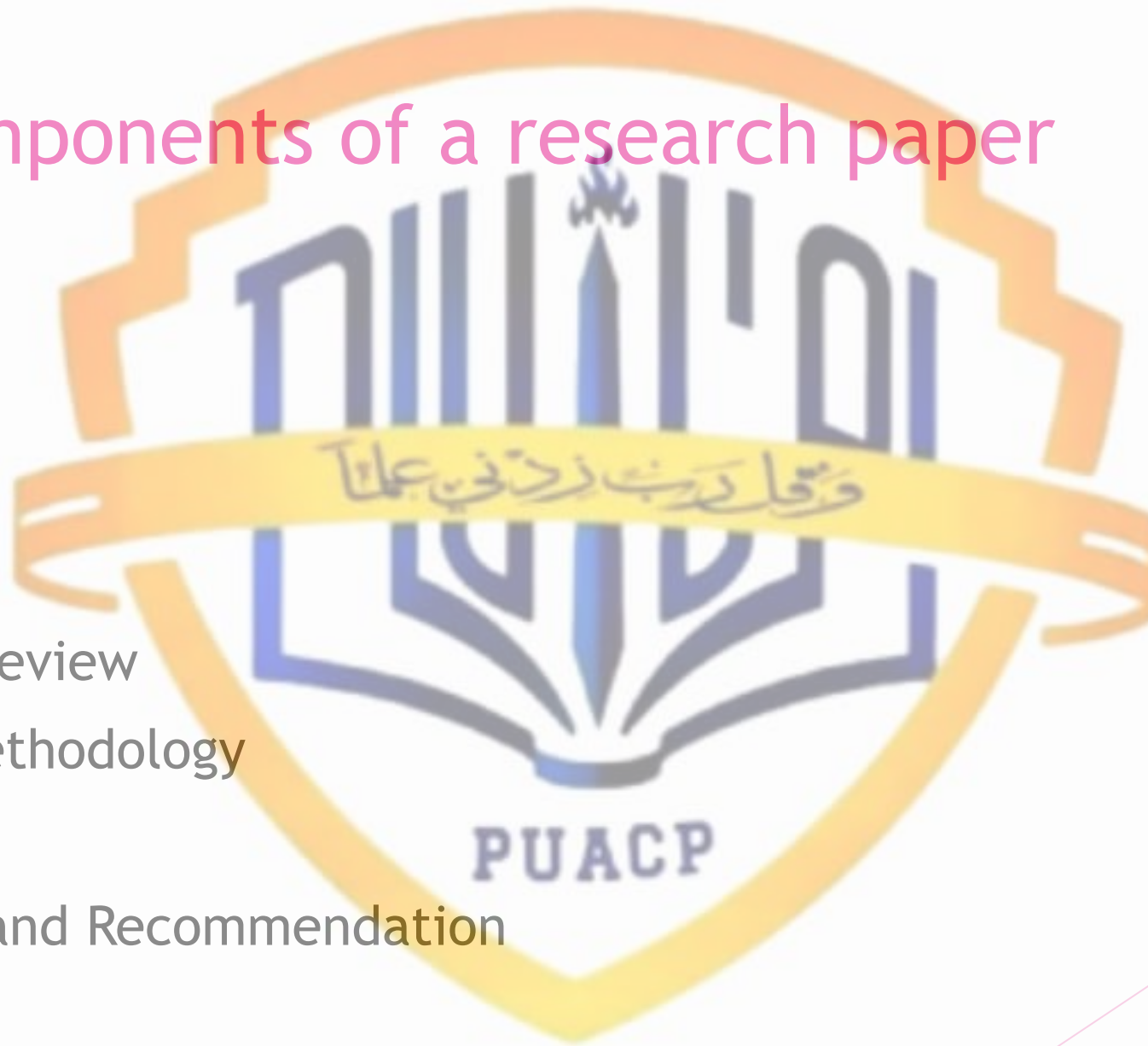
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Assistant Professor of Statistics

PUACP

Major components of a research paper

- ▶ Title
- ▶ Abstract
- ▶ Introduction
- ▶ Literature Review
- ▶ Data and Methodology
- ▶ Results
- ▶ Conclusion and Recommendation
- ▶ References



Title

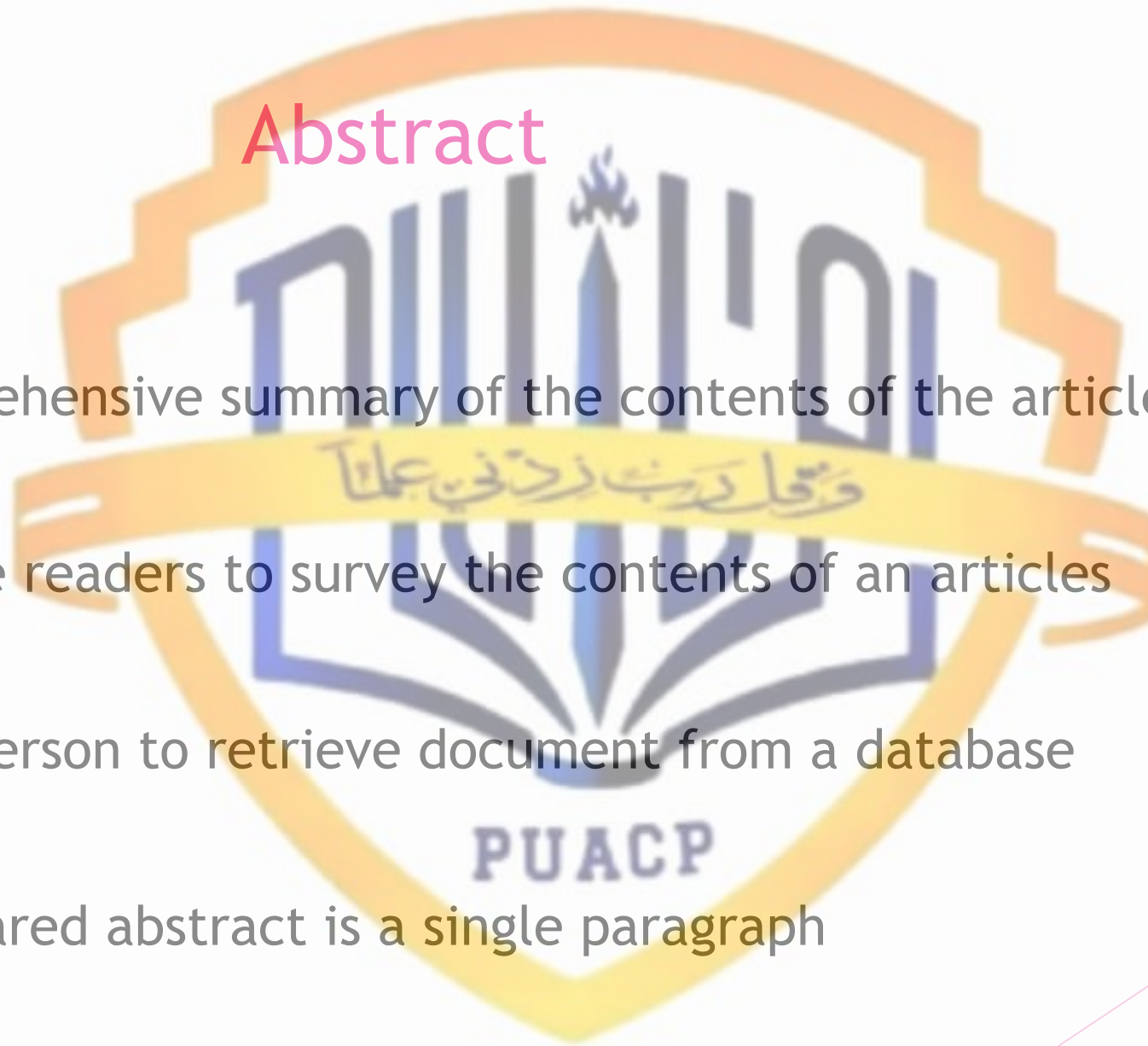
- ▶ Title should summarize the main idea of the manuscript
- ▶ It should be the concise statement of the main topic
- ▶ It should identify the variables and theoretical issues of subject
- ▶ It should be fully explanatory
- ▶ Its principle function is to inform the readers
- ▶ Avoid words that serves no useful purpose
- ▶ Title should be typed in upper case and lower case letters
- ▶ Centered between the left and right margin

Author's name and affiliation

- ▶ Author's name:
 - ▶ First- middle- last name
 - ▶ No use of initial
- ▶ Institutional Affiliation:
 - ▶ It identifies the location where the author or authors were when research was conducted
 - ▶ Include dual affiliation when two institutes contributed
 - ▶ If no affiliation, then city or state where he resides

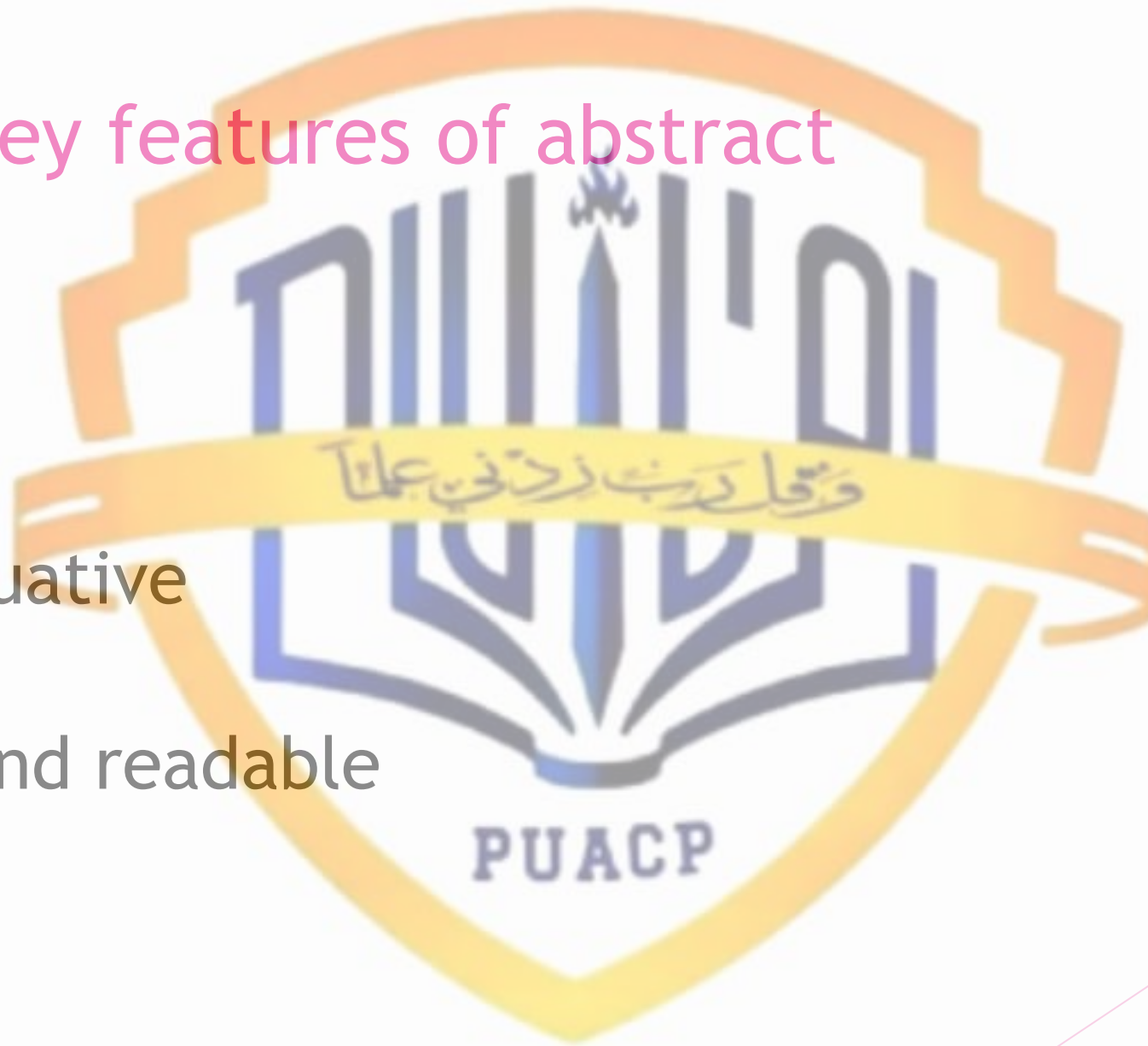
Abstract

- ▶ Brief comprehensive summary of the contents of the article
- ▶ It allows the readers to survey the contents of an articles
- ▶ It enables person to retrieve document from a database
- ▶ A well prepared abstract is a single paragraph



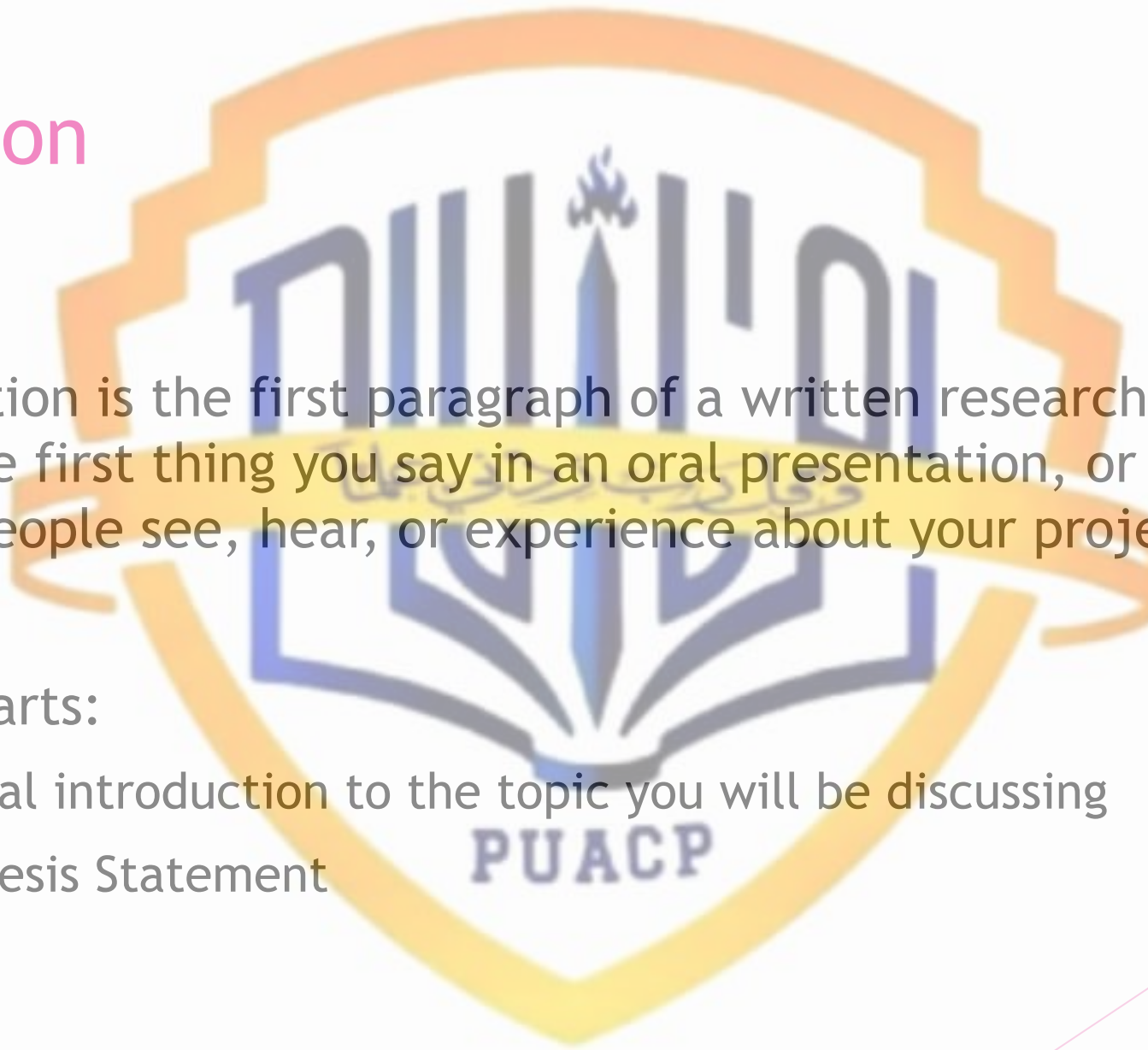
Key features of abstract

- ▶ Accurate
- ▶ Non evaluative
- ▶ Logical and readable
- ▶ Concise



Introduction

- ▶ An introduction is the first paragraph of a written research paper, or the first thing you say in an oral presentation, or the first thing people see, hear, or experience about your project.
- ▶ It has two parts:
 - ▶ 1. A general introduction to the topic you will be discussing
 - ▶ 2. Your Thesis Statement



Feature of Introduction

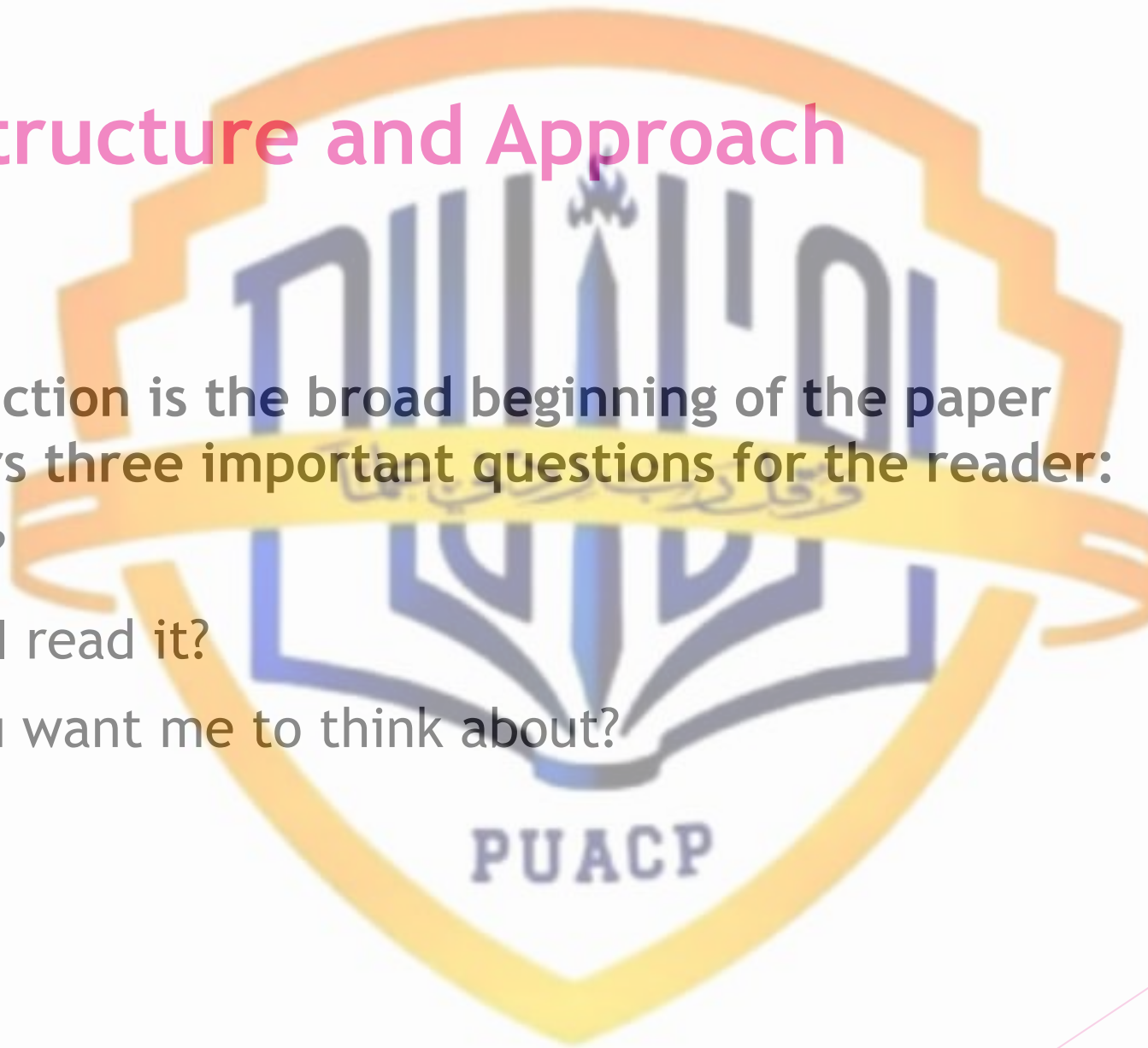
- ▶ Why is this problem important
- ▶ How does study relates to previous work in the research area
- ▶ What are the primary and secondary objectives
- ▶ What are the hypothesis
- ▶ What are the links with the theory
- ▶ How do the hypothesis and research design relate to each other
- ▶ What are the theoretical and practical implication of the study

When do I do it?

- ▶ Many books recommend writing your introduction last, after you finish your project.
- ▶ This is to make sure that you introduce what you are actually going to say.
- ▶ If your project changes in the creating process, it is important to make sure that your introduction accurately reflects what you will be saying.
- ▶ If you have written a good outline and stick to it, then it is fine to start writing your introduction first.
- ▶ Just make sure in your proofreading that you have kept the thread consistent throughout the paper.

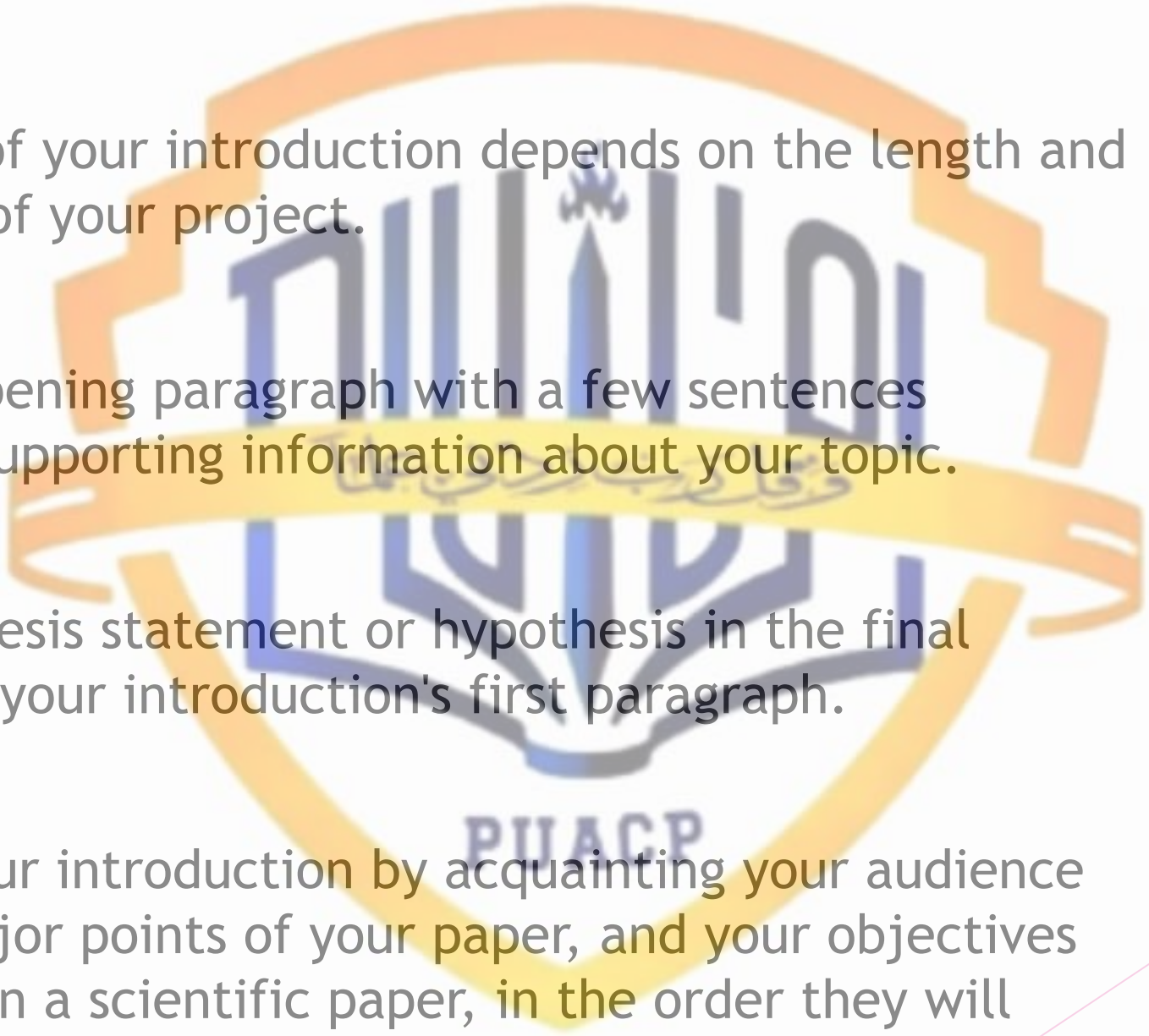
Structure and Approach

- ▶ The introduction is the broad beginning of the paper that answers three important questions for the reader:
- ▶ What is this?
- ▶ Why should I read it?
- ▶ What do you want me to think about?



How do I do it?

- ▶ Start with a couple of sentences that introduce your topic to your reader.
- ▶ You do not have to give too much detailed information; save that for the body of your paper.
- ▶ You do not have to give too much detailed information; save that for the body of your paper.
- ▶ Make these sentences as interesting as you can.
- ▶ Through them, you can hook a reader and get them very interested in the line of thinking you are going to develop in your project.
- ▶ state your thesis, which may be done in one or more sentences.

- 
- ▶ The length of your introduction depends on the length and complexity of your project.
 - ▶ Begin the opening paragraph with a few sentences containing supporting information about your topic.
 - ▶ Place the thesis statement or hypothesis in the final sentence of your introduction's first paragraph.
 - ▶ Continue your introduction by acquainting your audience with the major points of your paper, and your objectives and results in a scientific paper, in the order they will appear.

Literature Review

- ▶ A broad, comprehensive, in-depth, systematic, and critical review of scholarly publications, unpublished scholarly print materials, audiovisual materials, and personal communications
- ▶ Compare studies and know “*what’s out there*” on a specific topic
- ▶ Analytically examine existing body of research
- ▶ Bring order to an abundance of information spread across too many places

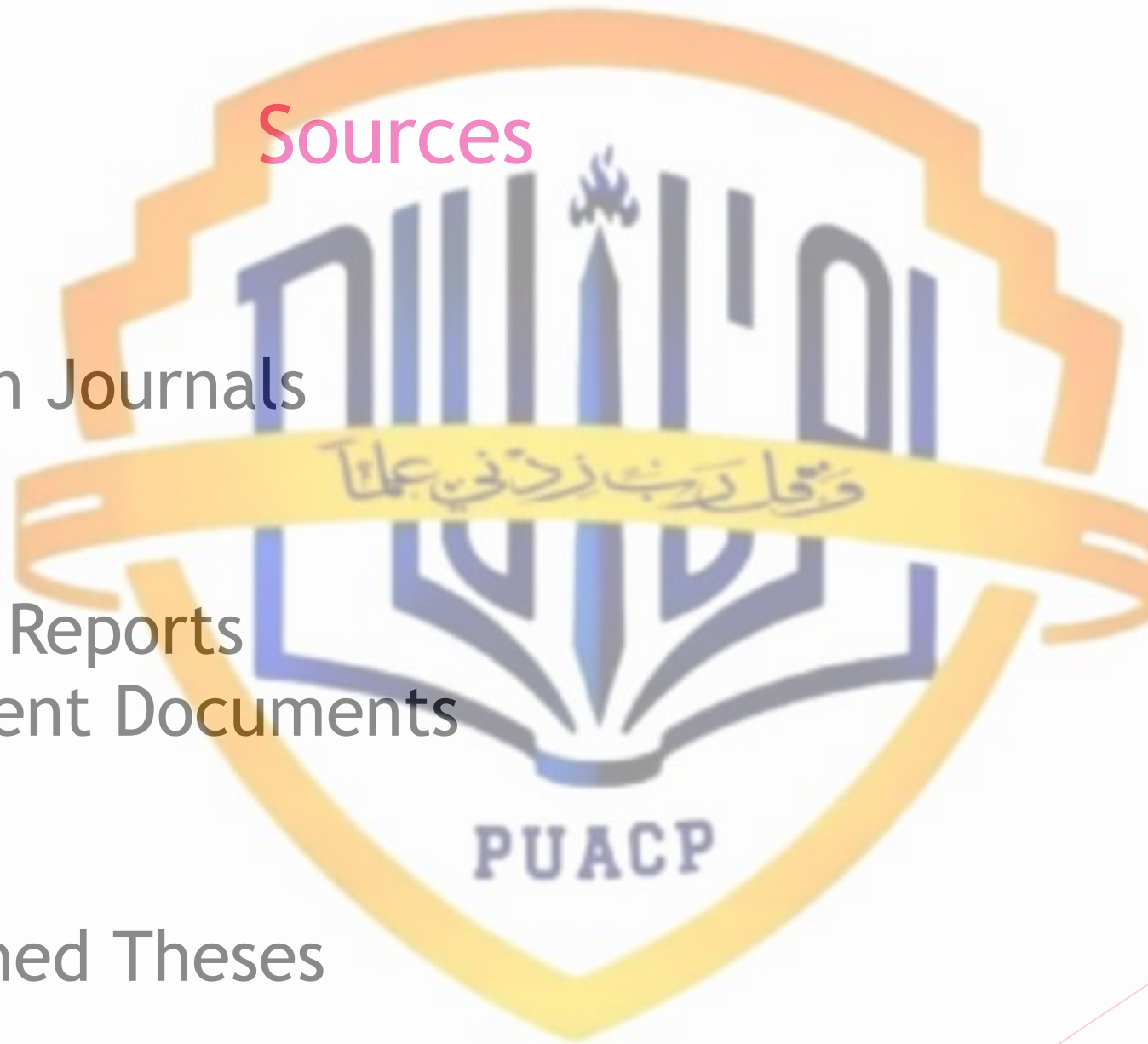
Purpose of a Literature Review

The literature review is a critical look at the existing research that is significant to the work that you are carrying out.

- ▶ To provide background information
- ▶ To establish importance
- ▶ To demonstrate familiarity
- ▶ To “carve out a space” for further research
- ▶ To make your research problem clear and bring focus into it
- ▶ To develop your methodology
- ▶ To know where you are
- ▶ To have a broader knowledge in your area of research

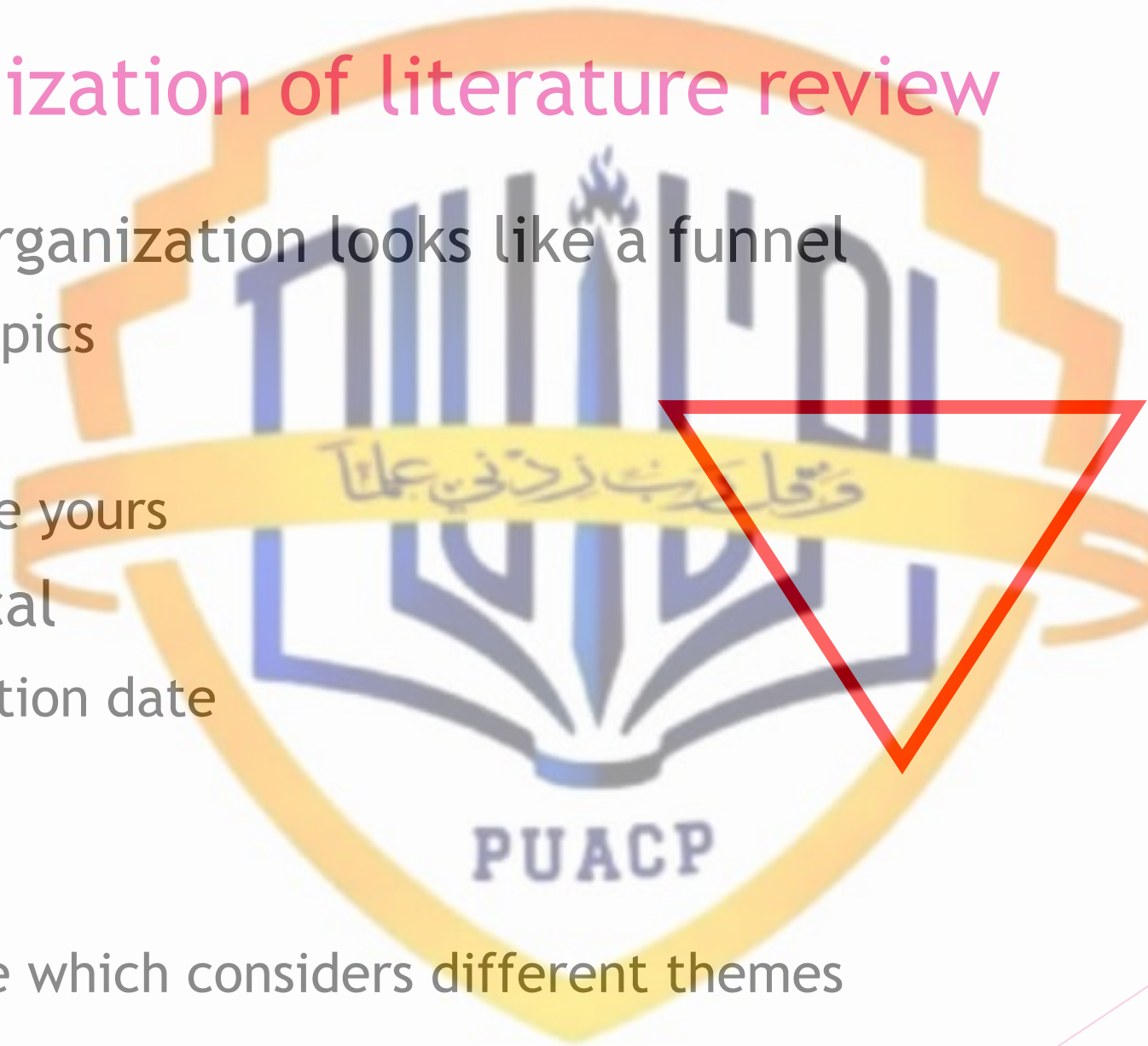
Sources

- ▶ Articles in Journals
- ▶ Books
- ▶ Internet
- ▶ Research Reports
- ▶ Government Documents
- ▶ Abstracts
- ▶ Reviews
- ▶ Unpublished Theses



Organization of literature review

- ▶ A general organization looks like a funnel
 - ▶ Broader topics
 - ▶ Subtopics
 - ▶ Studies like yours
- ▶ Chronological
 - ▶ By publication date
 - ▶ By trend
- ▶ Thematic
 - ▶ A structure which considers different themes

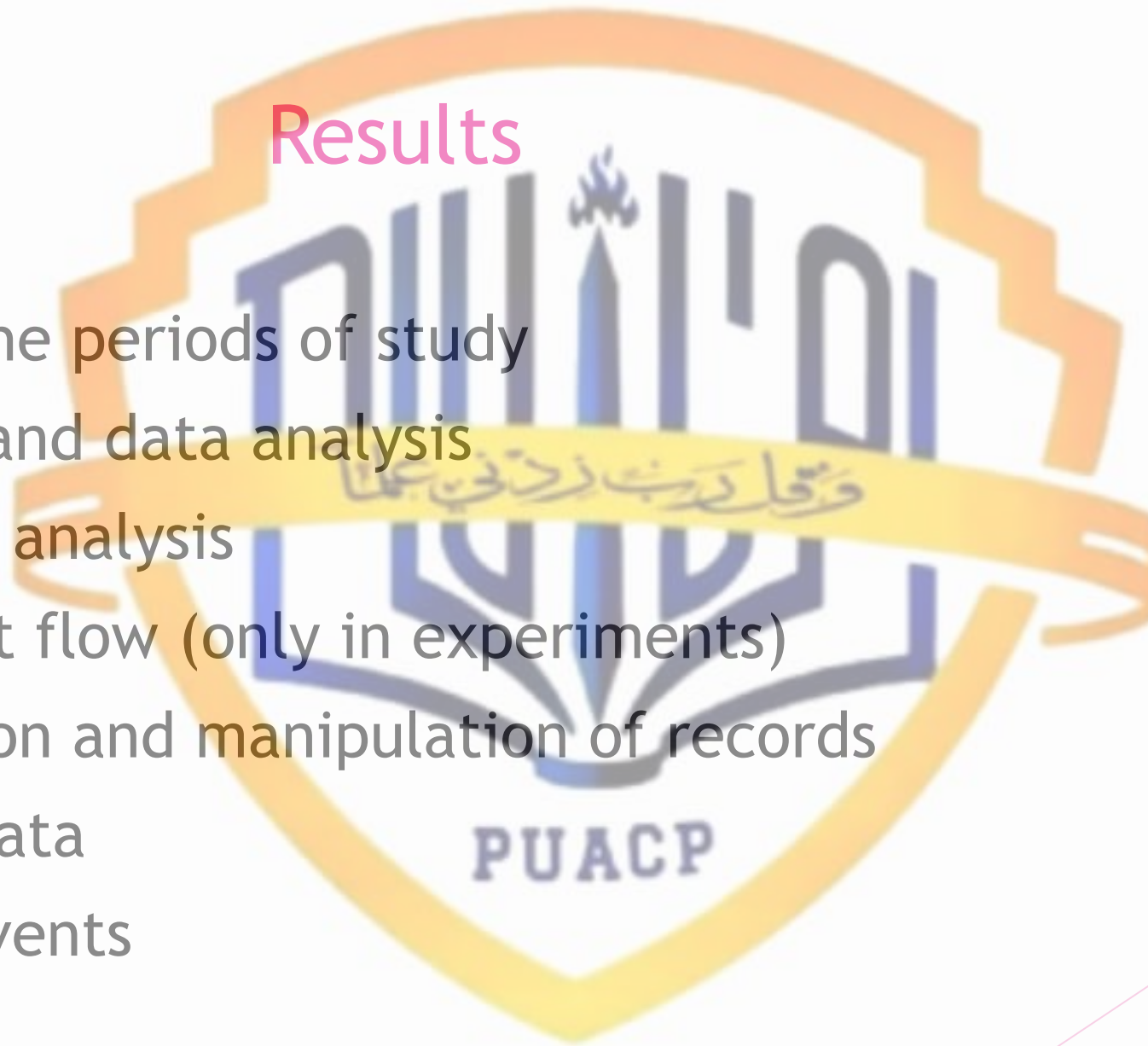


Methodology

- ▶ It describe in detail how the study was conducted
- ▶ It includes conceptual and operational definition of the variables used in the study
- ▶ Describing the materials used in the study
- ▶ Explaining how the materials were prepared
- ▶ Describing the research protocol
- ▶ Explaining how measurements were made and what calculations were performed
- ▶ Stating which statistical tests were done to analyze the data

Results

- ▶ Defining the periods of study
- ▶ Statistics and data analysis
- ▶ Secondary analysis
- ▶ Participant flow (only in experiments)
- ▶ Intervention and manipulation of records
- ▶ Baseline data
- ▶ Adverse events

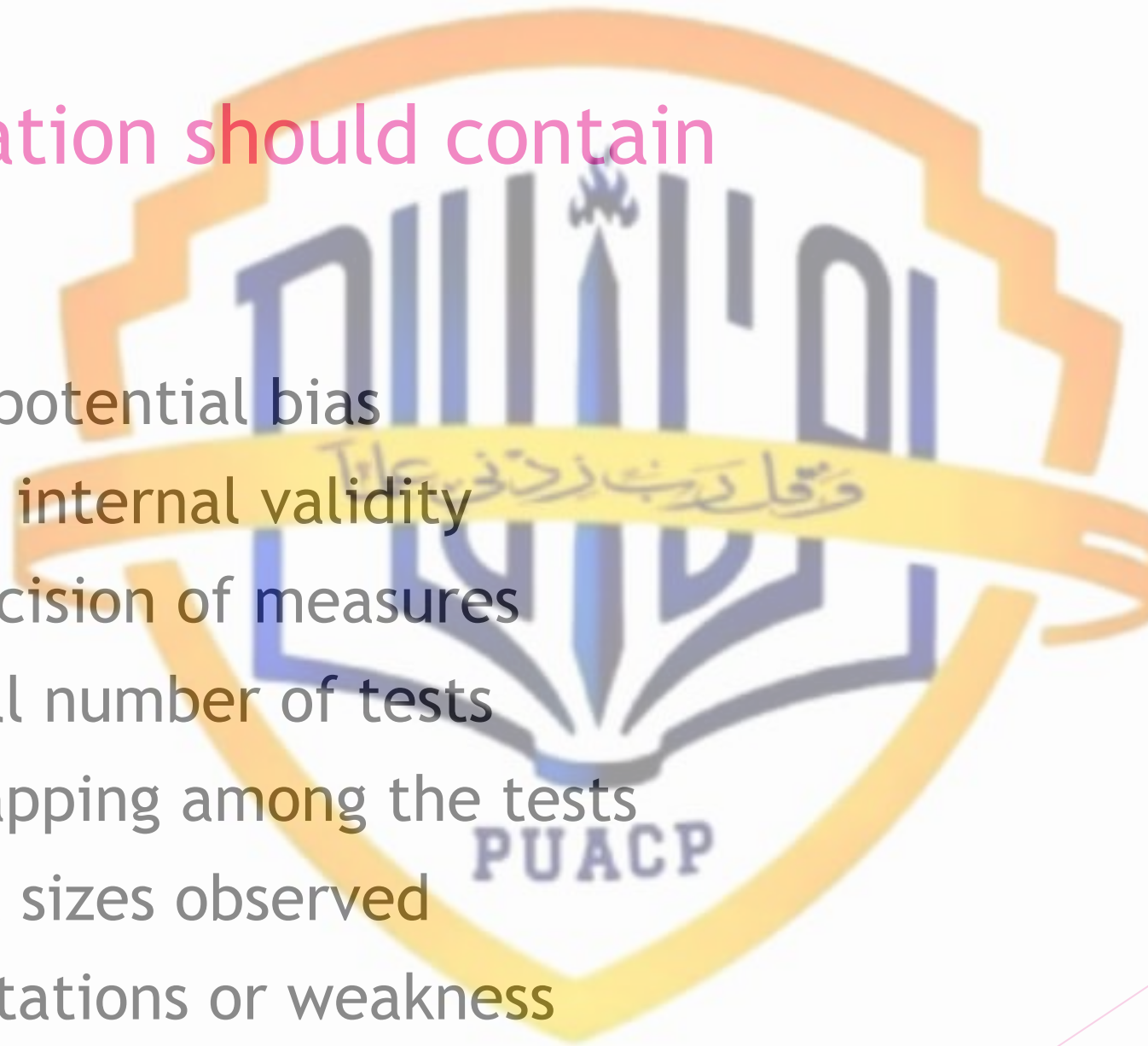


Discussions and conclusions

- ▶ Evaluate and interpret implication
- ▶ Draw inference and conclusions
- ▶ Description of the support or non support of the basic hypothesis of study
- ▶ Making inference about primary and secondary hypothesis
- ▶ If hypothesis are not supported then offer post-hoc explanation
- ▶ Similarities and differences between yours results and the results of the others

Interpretation should contain

- ▶ Source of potential bias
- ▶ Threats to internal validity
- ▶ The imprecision of measures
- ▶ The overall number of tests
- ▶ The overlapping among the tests
- ▶ The effect sizes observed
- ▶ Other limitations or weakness

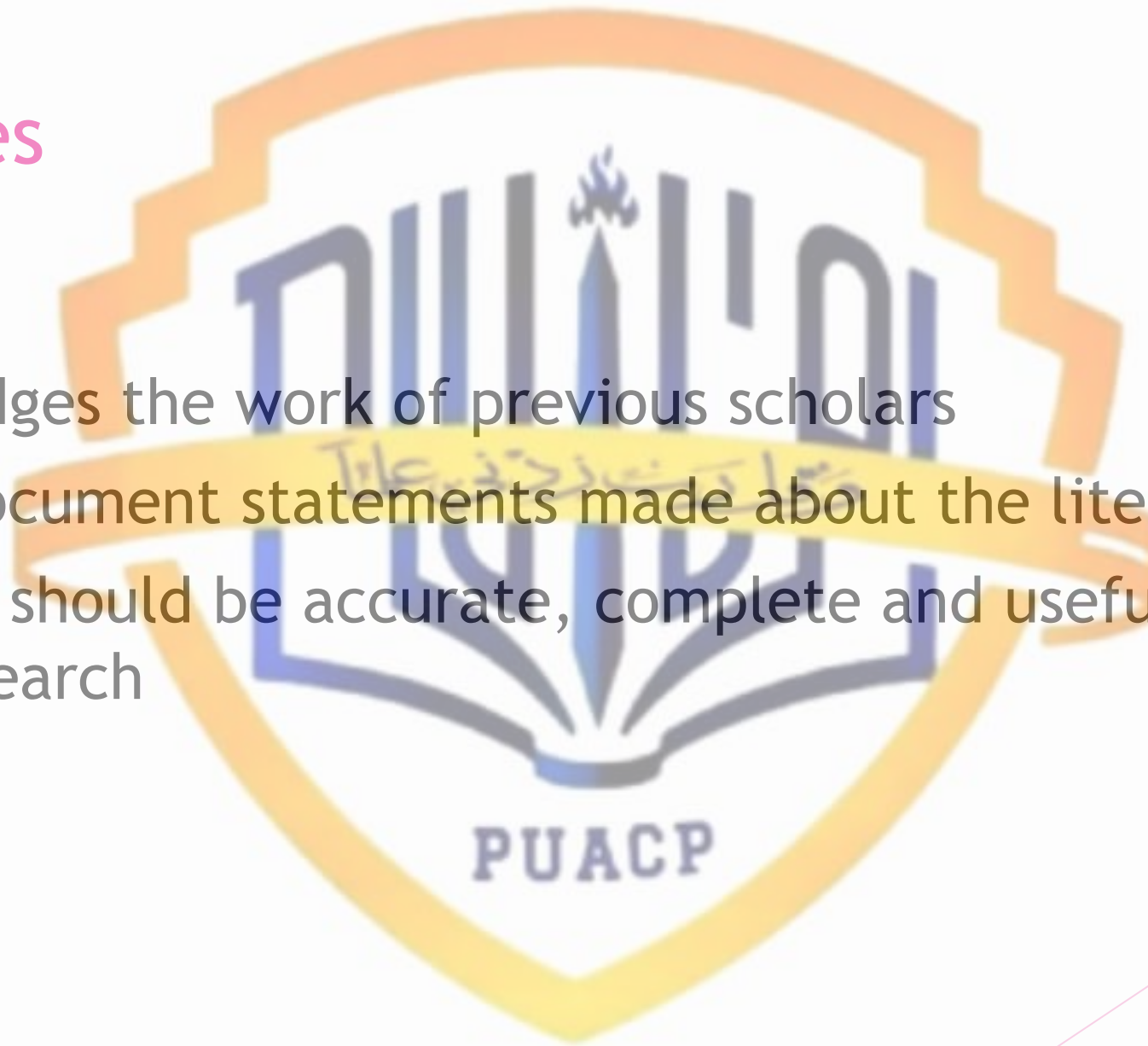


Limitations

- ▶ Limitations of your research
- ▶ Address alternative explanations of the results
- ▶ Discussion of the generalizability
- ▶ External validity of the findings
- ▶ Limitation between target population and the accessed population
- ▶ Possibilities that make more applications to new directions and their limitations
- ▶ What are the theoretical and practical limitation of the study
- ▶ What problems remained unresolved

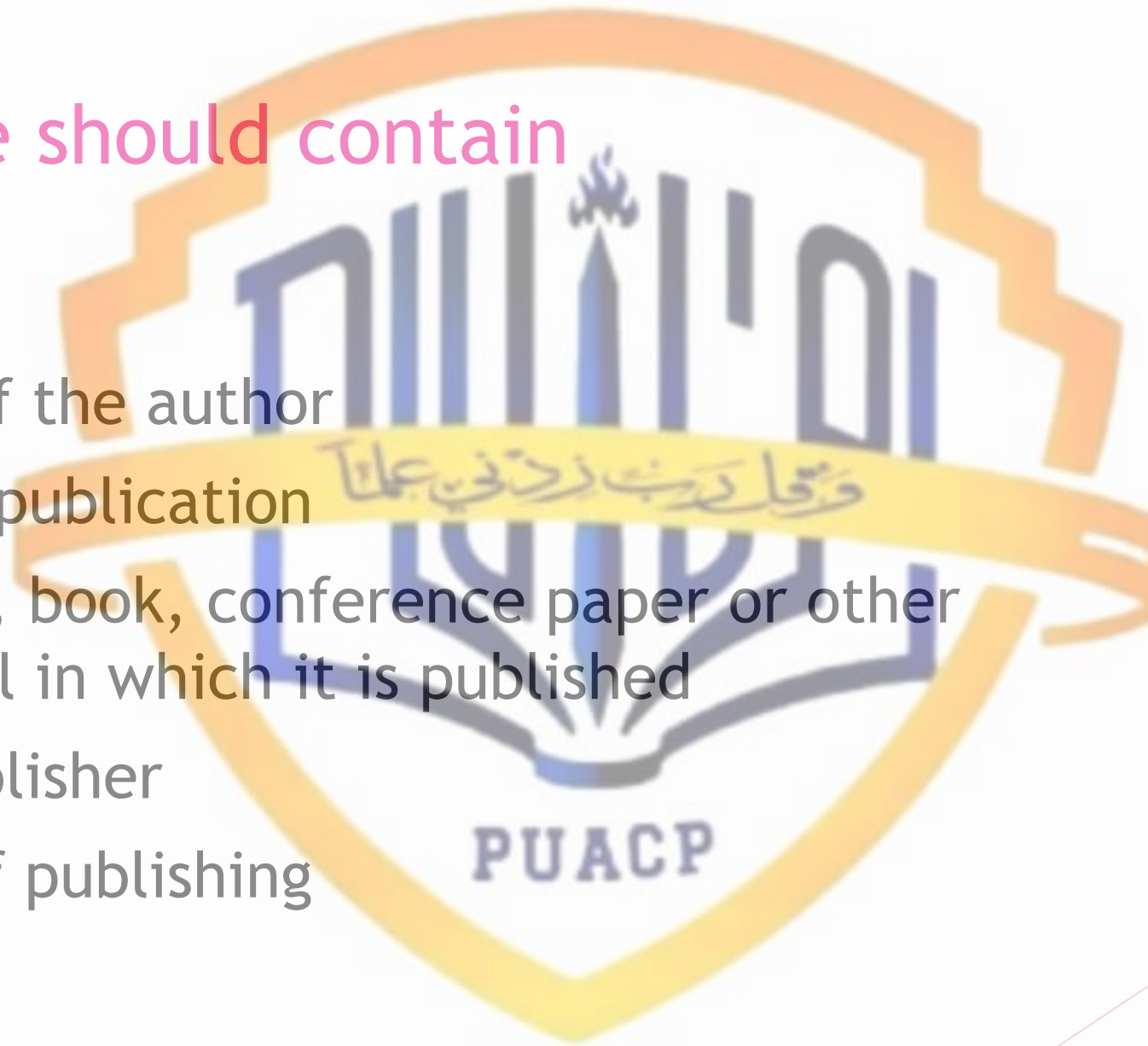
References

- ▶ Acknowledges the work of previous scholars
- ▶ Used to document statements made about the literature
- ▶ Reference should be accurate, complete and useful for future research



Reference should contain

- ▶ Name of the author
- ▶ Year of publication
- ▶ Journal, book, conference paper or other material in which it is published
- ▶ The publisher
- ▶ Place of publishing

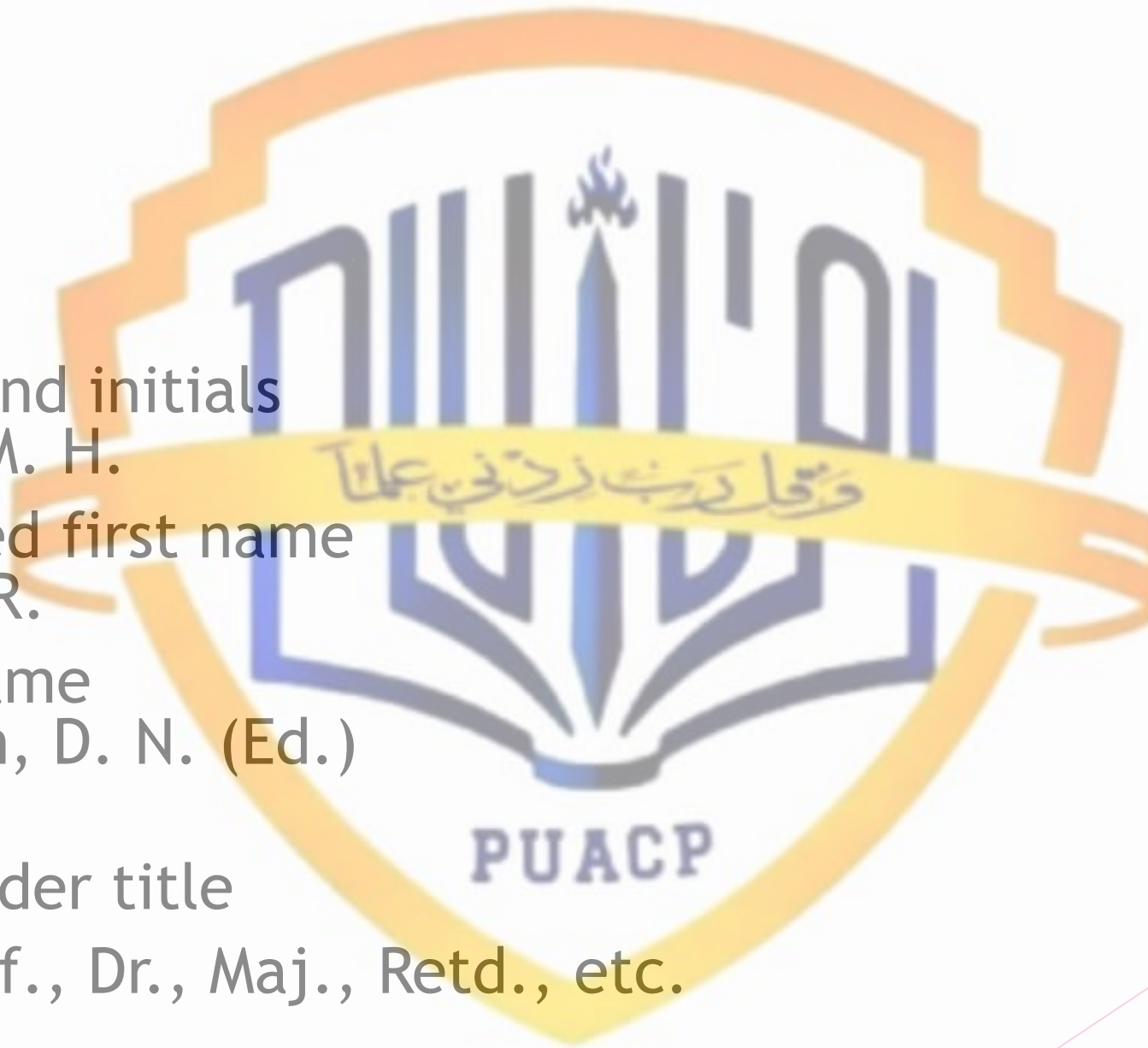


Various style manuals

- ▶ APA - American Psychological Association
- ▶ MLA - Modern Language Association
- ▶ Chicago Style - Chicago Manual of Style
- ▶ Turabian Style - based on Chicago Style
- ▶ Harvard Referencing System
- ▶ ASA - American Sociological Association
- ▶ CBE - Council of Biology Editors

Author

- ▶ Surname and initials
Kernis, M. H.
- ▶ Hyphenated first name
Sun, C.-R.
- ▶ Editor's name
Robinson, D. N. (Ed.)
- ▶ No author
Entry under title
- ▶ Delete Prof., Dr., Maj., Retd., etc.

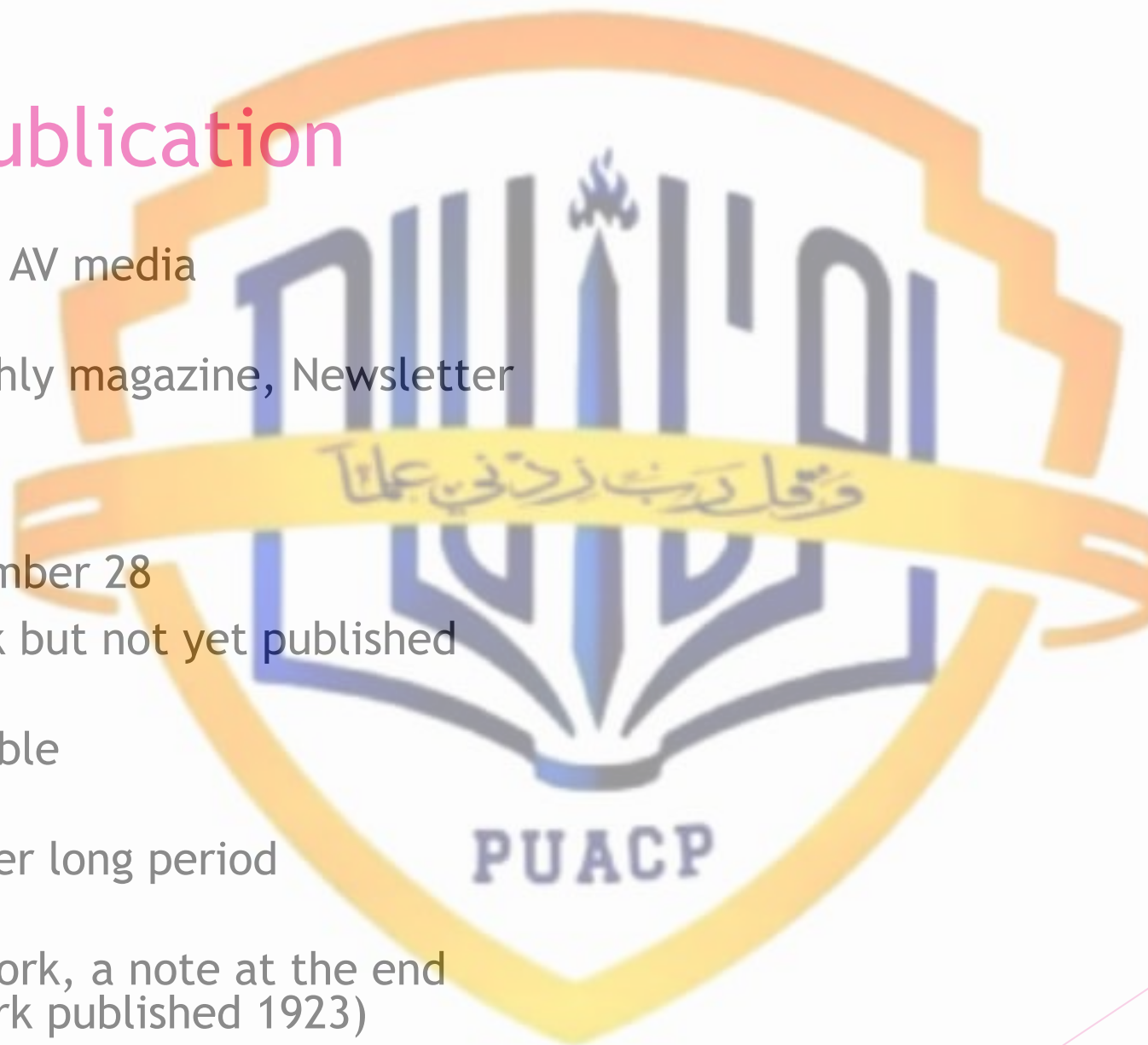


Authoring body or group

- ▶ Full name
National Institute of Health
- ▶ Subordinate body
University of the Punjab, Institute of
Business Administration
- ▶ Government agencies
Pakistan, Ministry of Finance

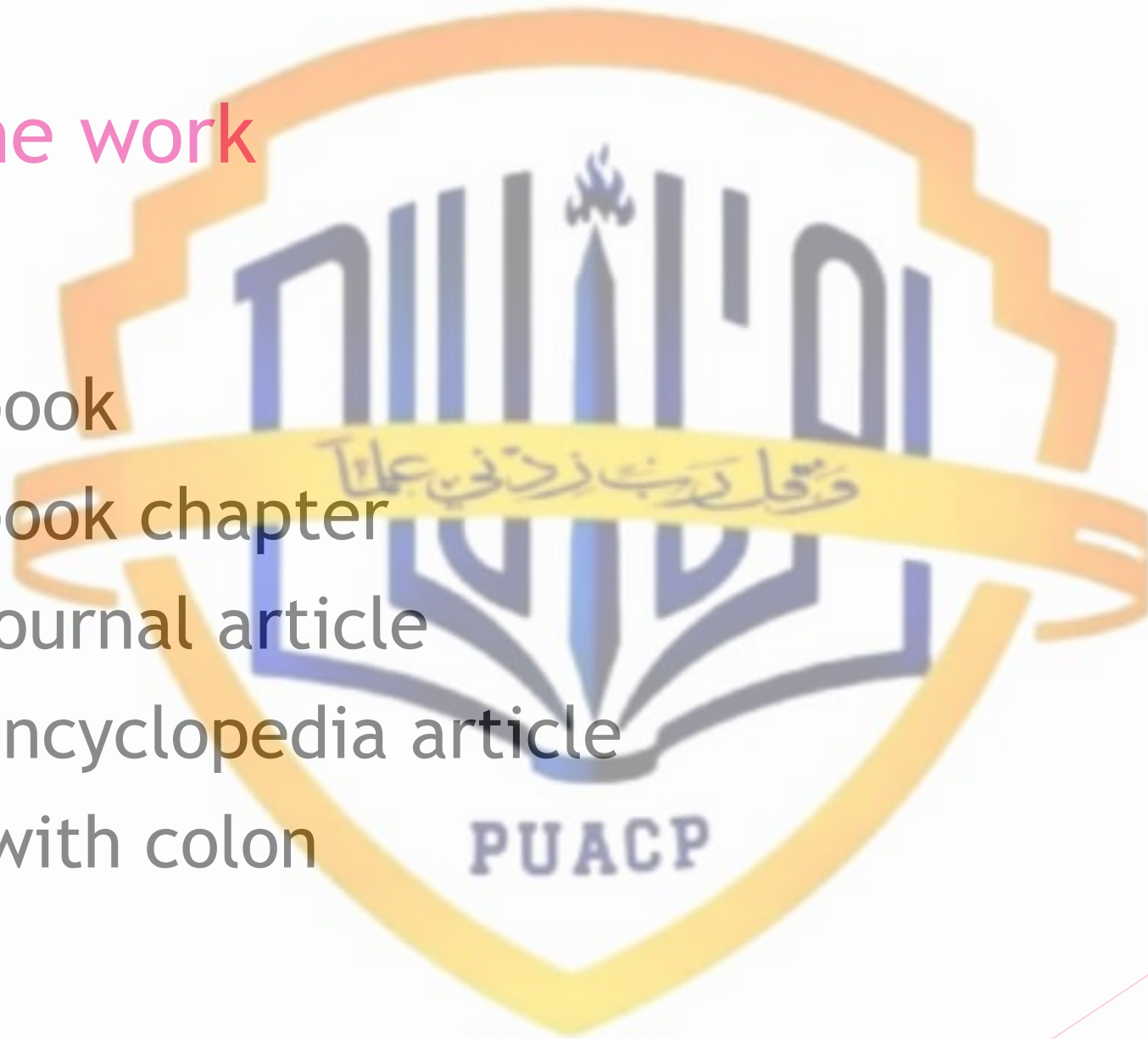
Date of publication

- ▶ Journal, book, AV media
1993
- ▶ Meeting, Monthly magazine, Newsletter
1993, June
1993, Spring
- ▶ Daily, Weekly
1994, September 28
- ▶ Accepted work but not yet published
in press
- ▶ No date available
n.d.
- ▶ Publication over long period
1959-1963
- ▶ Republished work, a note at the end
(Original work published 1923)



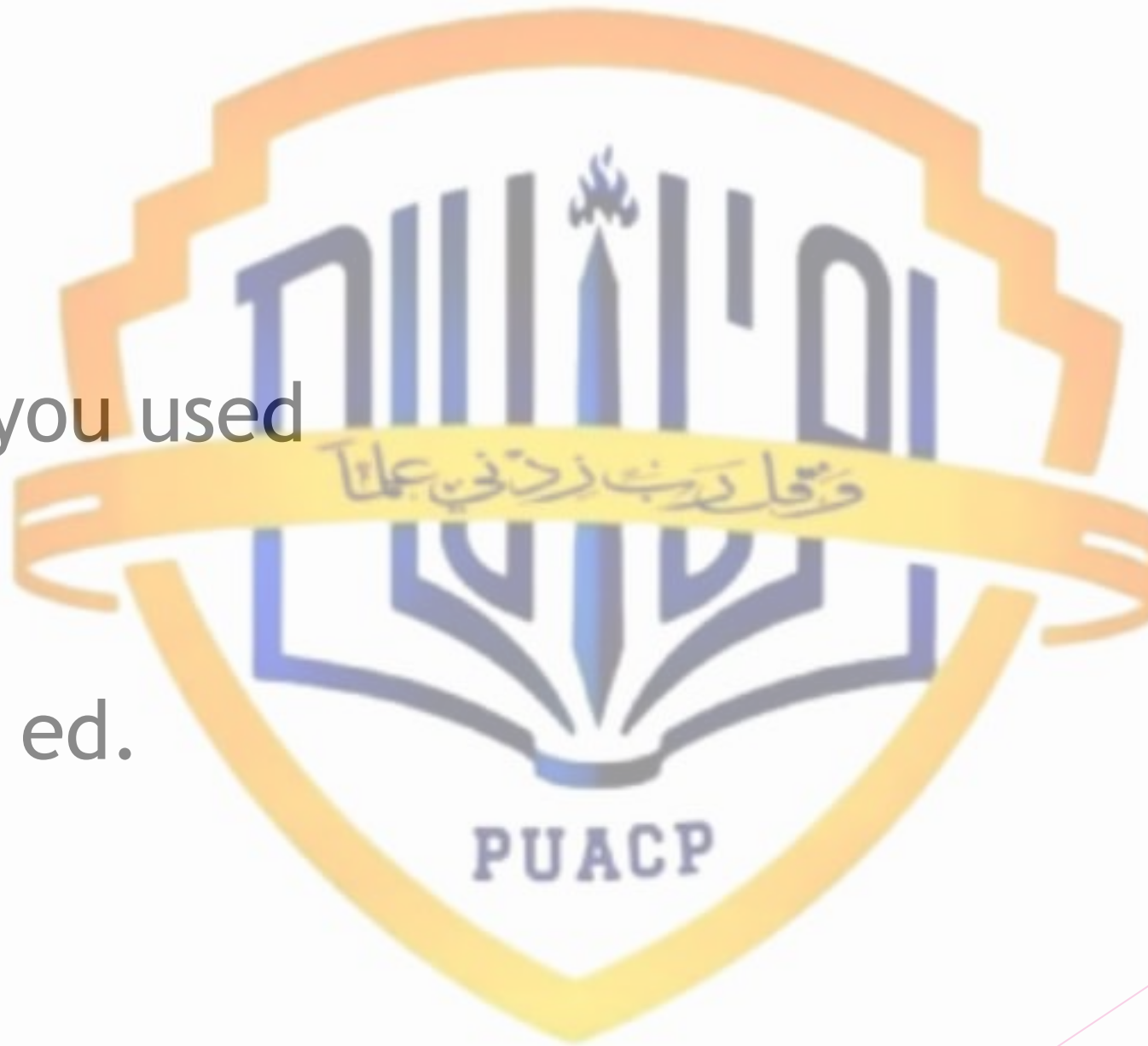
Title of the work

- ▶ Title of book
- ▶ Title of book chapter
- ▶ Title of journal article
- ▶ Title of encyclopedia article
- ▶ Subtitle with colon



Edition

- Edition you used
- 2nd ed.
- Rev. ed.
- 4th rev. ed.



Publisher

- ▶ Publisher name for non periodicals
- ▶ In a brief form
- ▶ Omit superfluous terms, such as Publishers, Publications, Co., Inc.
 - Sage
 - Wiley
 - McGraw-Hill
 - Prentice Hall
 - Ferozsons
- ▶ Use only word “Author” when author and publisher is the same

Place of publication

- ▶ Name of city
- ▶ If city is not well known then add state/province and/or country
Jaipur, India
Medford, NJ
- ▶ US postal service abbreviations for states (2-digit codes)
CA for California
- ▶ If more cities are given, use the first or the publisher's head office if clearly mentioned

Title of the source

- ▶ Title of the book in case of a book chapter
- ▶ Title of the journal in case of journal article
- ▶ Journal title in full
Harvard Business Review
No Har. Bus. Rev.
No HBR

Location

- ▶ Journal volume and issue number in numerals 33(4)
- ▶ Volume of a book
Vols. 1-20
(Vol.26, pp. 501-508)
- ▶ Start and end (inclusive) page numbers for journal article or book chapter
215-224
(pp. 215-224)
- ▶ Discontinuous pages
5-7, 11-12

URL or DOI

- ▶ Uniform Resource Locator (URL)

<http://www.topicsinclinicalnutrition.com>

- ▶ Digital Object Identifier (DOI)

10.1037/0002-9432.76.4.482





Citing in text

Author's name

- ▶ in sentence

- ▶ Schwepps (1998) states that the solution sat dormant for several months before any of the employees tested it (p. 743).

- ▶ in parentheses

- ▶ When the solution had been sitting for a number of months, the employees tested for bacteria (Schwepps, 1998).

Short quotation

- ▶ When fewer than 40 words
- ▶ Put prose quotation in running text
- ▶ Put quote marks around quoted material
- ▶ Author's last name, publication year, and page number(s) of quote must appear in the text

Example - Short quotation

Caruth (1996) states that a traumatic response frequently entails a “delayed, uncontrolled repetitive appearance of hallucinations and other intrusive phenomena” (p. 11).

A traumatic response frequently entails a “delayed, uncontrolled repetitive appearance of hallucinations and other intrusive phenomena” (Caruth, 1996, p. 11).

Long quotations

- ▶ When 40 words or more
- ▶ In block form
- ▶ Indent 5-7 spaces and omit the quotation marks. If the quotation has internal paragraphs, indent the internal paragraphs a further 5-7 spaces
- ▶ Do not use quotation marks
- ▶ Double space the block quote
- ▶ Cite the source after the end punctuation of the quote

Example - Long quotation

Meile (1993) found the following:

The “placebo effect,” which had been verified in previous studies, disappeared when behaviors were studied in this manner. Furthermore, the behaviors were never exhibited again, even when real drugs were administered. Earlier studies were clearly premature in attributing the results to a placebo effect. (p. 276)

Multiple authors

- ▶ 2 authors - cite both names separated by &
Example:
(Kosik & Martin, 1999, p. 127)
- ▶ 3-5 authors - cite all authors first time; after first time, use et al.
Example:
(Wilson et al., 2000)

Multiple citations

- ▶ Multiple sources from same author - chronological order, separated by comma
(Burke, 1998, 1999, in press)
- ▶ Within same year:
(Burke, 1998a, 1998b, 1999, in press)
- ▶ Multiple sources - separated by semicolon, alphabetical order
(Burke, 1998; Perez, 1992; Wilhite, 2001)

Handling parenthetical citations

- ▶ More than one author with the same last name
(H. James, 1878); (W. James, 1880)
- ▶ Specific part of a source
(Jones, 1995, chap. 2)

Sample parenthetical citations

Recently, the history of warfare has been significantly revised by Higdon et al (1987), Marcus (1989), and Raitt and Tate (1997) to include women's personal and cultural responses to battle and its resultant traumatic effects. Feminist researchers now concur that "It is no longer true to claim that women's responses to the war have been ignored" (Raitt & Tate, p. 2). Though these studies focus solely on women's experiences, they err by collectively perpetuating the masculine-centered impressions originating in Fussell (1975) and Bergonzi (1996).

However, Tylee (1990) further criticizes Fussell, arguing that his study "treated memory and culture as if they belonged to a sphere beyond the existence of individuals or the control of institutions" (p. 6).

Reference List / Bibliography



Reference list

- ▶ Place the list of references cited at the end of the paper
- ▶ Start references on a new page
- ▶ Begin each entry flush with the left margin
- ▶ Indent subsequent lines five to seven spaces (hanging indent)
- ▶ Double space both within and between entries
- ▶ *Italicize* the title of books, magazines, etc.

Reference list order

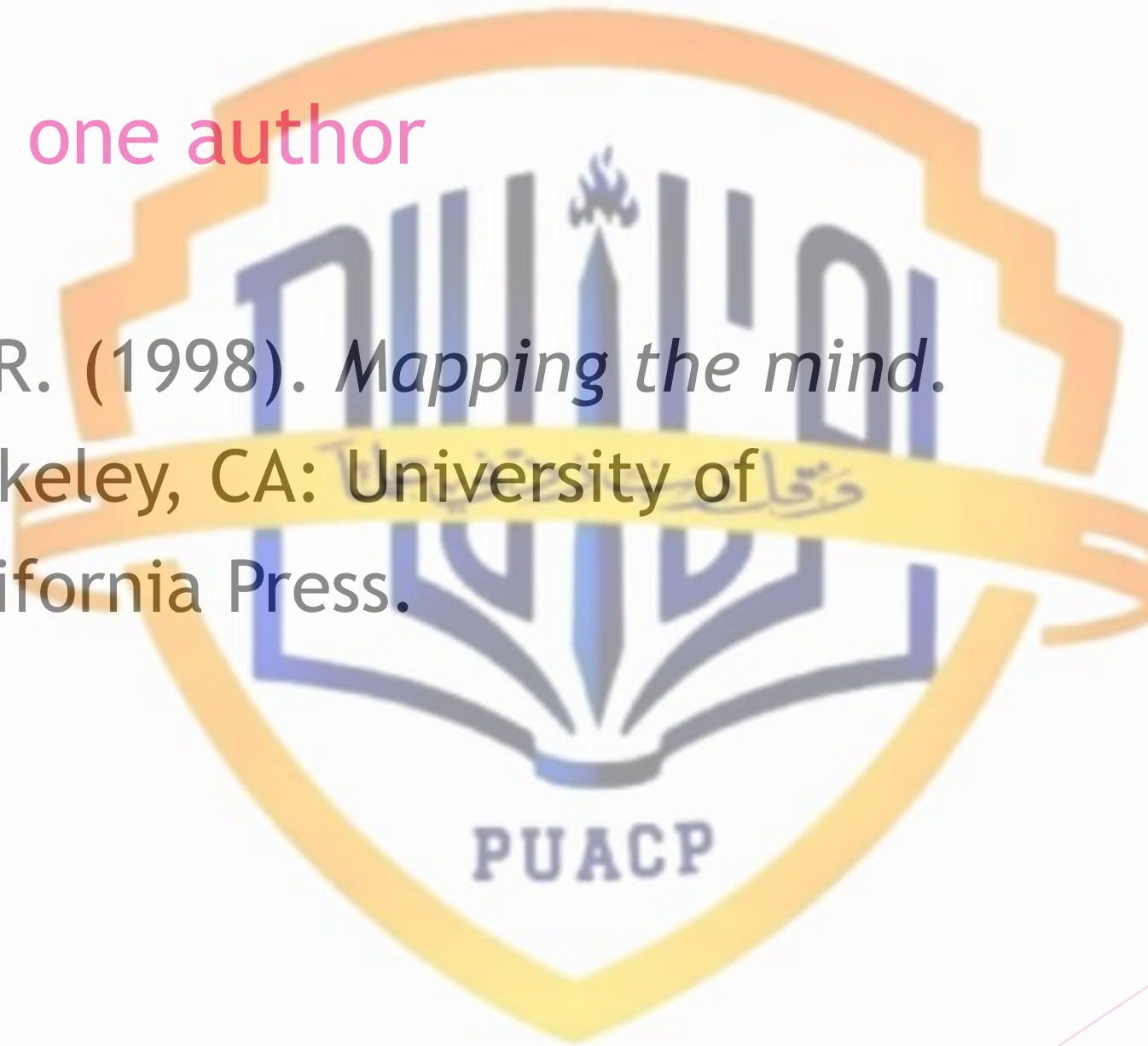
- ▶ Arrange sources alphabetically beginning with author's last name
- ▶ If author has more than one source, arrange entries by year, earliest first
- ▶ When an author appears both as a sole author and, in another citation as the first author of a group, list the one author entries first
- ▶ If no author given, begin entry with the title and alphabetize without counting a, an, or the
- ▶ Do not underline, italicize or use quote marks for titles used instead of an author name

Example - Reference list order

- ▶ Baheti, J. R. (2001a). Control ...
- ▶ Baheti, J. R. (2001b). Roles of ...
- ▶ Kumpfer, K. L. (1999). Factors ...
- ▶ Kumpfer, K. L. (2002). Prevention ...
- ▶ Kumpfer, K. L., Alvarado, R., Smith, P., ...
- ▶ Yoshikawa, H. (1994). Preventions ...

Book with one author

Carter, R. (1998). *Mapping the mind*.
Berkeley, CA: University of
California Press.



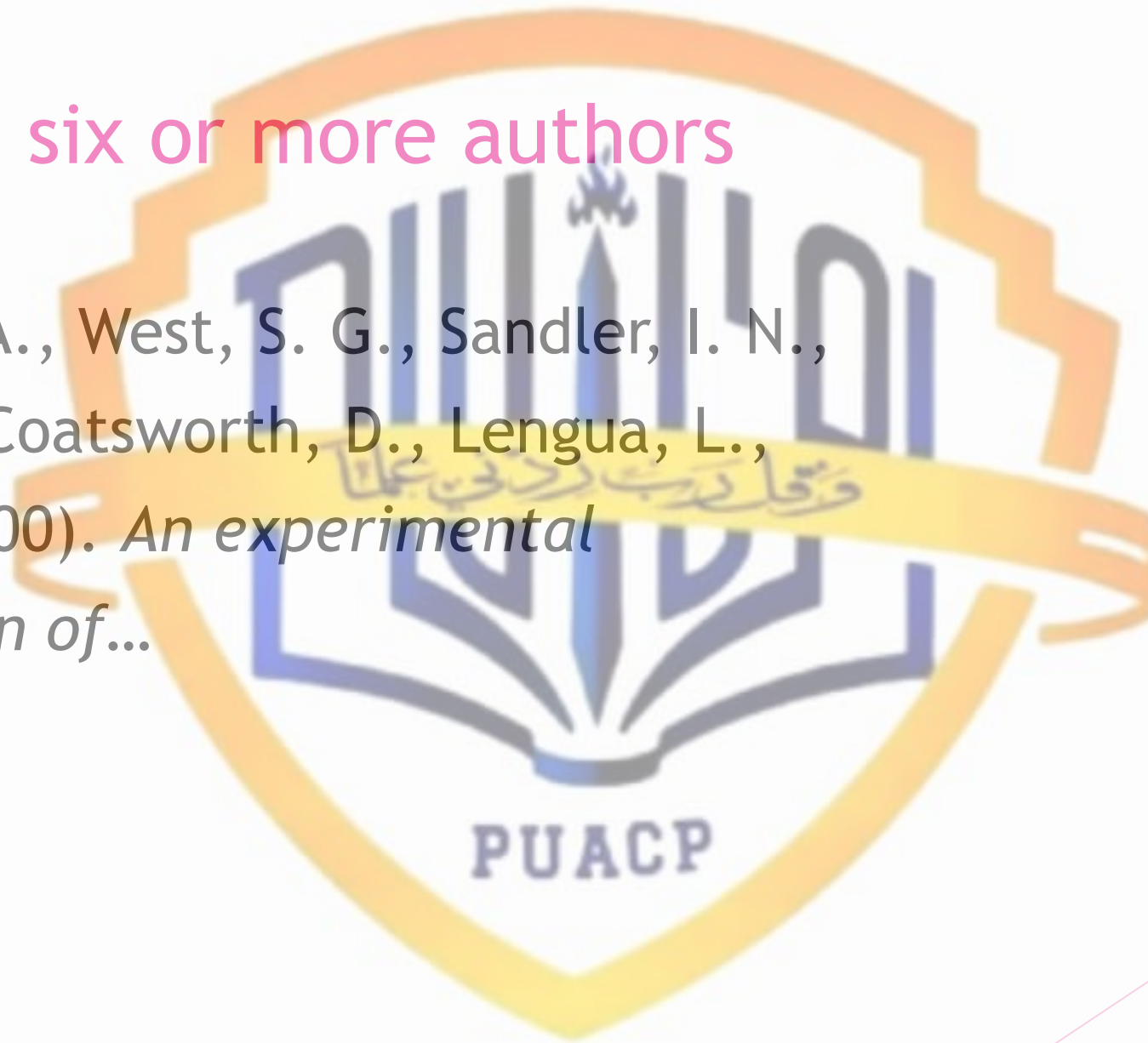
Book with two authors

Struck, W., & White, E. B. (1979).
The elements of style (3rd ed.).
New York: Macmillan.



Book with six or more authors

Wolchik, S. A., West, S. G., Sandler, I. N.,
Tein, J., Coatsworth, D., Lengua, L.,
et al. (2000). *An experimental
evaluation of...*



Book with no author

Merriam-Webster's collegiate dictionary
(10th ed.). (1993). Springfield, MA:
Merriam-Webster.

Newspaper

Schwartz, J. (1993, September 30).
Obesity affects economic, social
status. *The Washington Post*, pp.
A1, A4.

Thesis

Ho, M. (2000). *Coping strategies of counseling professionals*
(Unpublished master's thesis,
Nanyang Technological University,
Singapore).



Web page

Green, C. (2000, April 16). *History & philosophy of psychology web resources*. Retrieved from <http://www.yorku.ca/dept.htm>

Sample Reference List

References

- Calvillo, D. (1999). *The theoretical development of aggression*. Retrieved August 21, 2002 from: <http://www.csubak.edu/~1vega/dustin2.html>
- Flory, R. K., (1969a). Attack behavior as a function of minimum inter-food interval. *Journal of the Experimental Analysis of Behavior*. 12, 825-828.
- Flory, R. K., (1969b). Attack behavior in a multiple fixed-ratio schedule of reinforcement. *Psychonomic Science*, 16, 383-386.
- Flory, R. K., & Everist, H.D. (1977). The effect of a response requirement on schedule- induced aggression. *Bulletin of the Psychonomic Society* 9, 383-386.
- Gentry, W. D. (1968). Fixed-ratio schedule-induced aggression. *Journal of the Experimental Analysis of Behavior* 11, 813-817.

Plegarism

- Plagiarism is the use of the words and/or ideas of another person without acknowledging the source.



To avoid plagiarism,

- ▶ Acknowledge your sources with in-text citations and a reference page.
- ▶ Enclose direct quotations in quotation marks or otherwise indent them from the body of your text.
- ▶ If you use another person's idea or paraphrase another person's words, be sure to use your own language and style of writing – don't simply rearrange the words.
- ▶ Use an in-text citation to acknowledge the source, then list on a reference page the publications or sources from which you obtained your citations.

THANKS

